

Machine 3: A Multiplicative-Coincidence Cryptid

The $A(a, b)$ system — halting reduced to hitting an exact power of 3

1. Overview

Machine 3 is the deterministic system $A(a, b)$ on positive-integer pairs defined by six guarded rules (Section 2), started at $A(1, 1)$. It is the third machine in this collection, and it fills a gap the meta report predicted would eventually appear: a **multiplicative-coincidence** cryptid, the type of the bbchallenge machine Space Needle, where halting requires the orbit to land on an exact power rather than on a sparse affine family (machine 1) or to trip a cumulative walk (the Hydra family).

The analysis is clean and largely complete except for the open core. **Halting is equivalent to a single, sharply stated event** (Theorem 1): the a -coordinate must reach an exact power of 3 whose exponent is divisible by 3. The divide-chains that dominate the running time have a closed form (Theorem 2); an affine potential proves the machine cannot cycle (Theorem 3) — another machine in the collection to yield one, by the same recipe; and the branch statistic is the base-3 valuation law, the exact base-3 analogue of Collatz. Verified: no halt in 3,000,000 composite steps — the a -coordinate reaching 708,337 bits over 2,249,864 resets, and the orbit never once landing on a power of 3 (no $a = 1$ event at all). The machine almost certainly never halts, for the Space-Needle reason — but a proof would require deciding whether one orbit ever hits a power, which is exactly the open Collatz-type obstacle.

2. The machine

State $A(a, b)$, $a \geq 1$, $b \geq 1$. The rules (first match; $k \geq 1$ and $b \geq 1$ where written):

#	rule	guard	role
R1	$A(1, 3k) \rightarrow \text{halt}$	$k \geq 1$	b divisible by 3
R2	$A(1, 3k+1) \rightarrow A(3k+4, 1)$	—	$b \equiv 1 \pmod{3}$
R3	$A(1, 3k+2) \rightarrow A(3k+3, 2)$	—	$b \equiv 2 \pmod{3}$
R4	$A(3k, b) \rightarrow A(k, b + 2k + 1)$	$k \geq 1$	$a \equiv 0 \pmod{3}$: divide
R5	$A(3k+1, b) \rightarrow A(4k + b + 3, 1)$	$k \geq 1$	$a \equiv 1 \pmod{3}$: reset
R6	$A(3k+2, b) \rightarrow A(4k + b + 5, 1)$	—	$a \equiv 2 \pmod{3}$: reset

Dispatch: $a \equiv 1$ uses R1–R3 (on $b \pmod{3}$); $a \geq 2$ uses R4–R6 (on $a \pmod{3}$). Two behaviors alternate. When $a \not\equiv 0 \pmod{3}$ the **reset** rules R5/R6 map a to roughly $4a/3$, adding the current b and setting b back to 1. When $a \equiv 0 \pmod{3}$ the **divide** rule R4 replaces a by $a/3$ and pumps b upward. Halting is possible only from $a = 1$, and only when $b \equiv 0 \pmod{3}$.

Fidelity. The implementation reproduces the hand-computed trajectory $A(1,1) \rightarrow A(4,1) \rightarrow A(8,1) \rightarrow A(14,1) \rightarrow \dots$, and the accelerated form (Section 4) matches the base machine state-for-state on the first 177,717 composite steps of the real orbit before it outruns the base horizon.

3. The halting criterion

The gateway to halting is $a = 1$, and $a = 1$ is reached only by the divide rule R4 grinding a down. R4 divides a by 3, so it reaches 1 **only from an exact power of 3**: if $a = 3^j \cdot M$ with M not divisible by 3, the divide chain stops at M , and $M = 1$ iff a was a pure power 3^j . Section 4 (Theorem 2) gives the value of b there: $b = 3^j + j$. Since $3^j \equiv 0 \pmod{3}$ for $j \geq 1$, $b \equiv j \pmod{3}$, and R1 halts iff $b \equiv 0 \pmod{3}$. Hence:

Theorem 1. Started at $A(1, 1)$, the machine halts **if and only if** the a -coordinate reaches an exact power of 3 whose exponent is divisible by 3:

$$a \in \{3^j : j \equiv 0 \pmod{3}\} = \{27, 729, 19683, \dots\} = \{27^m : m \geq 1\}.$$

(Verified: for $j = 1..15$, placing the machine at $(3^j, 1)$ halts exactly when $j \equiv 0 \pmod{3}$.) This is the multiplicative analogue of machine 1's affine halting family: the target is a **geometric set of exact powers**, spaced by a factor of 27, precisely the shape of Space Needle's power-of-2 target on the bbchallenge wiki — but with an added congruence ($\text{exponent} \equiv 0 \pmod{3}$) selecting one third of the powers.

4. Acceleration and the potential

4.1 Theorem 2: the divide-chain closed form

The divide chain is the only expensive event. From (N, b_0) with $N = 3^j \cdot M$, $M \not\equiv 0 \pmod{3}$, applying R4 j times gives

$$(N, b_0) \rightarrow (M, b_0 + (N - M) + j),$$

since level i adds $2(N/3^i) + 1$ and the geometric sum telescopes to $(N - M) + j$. This collapses an $O(N)$ -long chain to one jump (30,000 randomized checks). With it, the machine simulates to millions of composite steps in seconds. When $M = 1$ the state is $A(1, N - 1 + j) = A(1, 3^j + j)$, which is where Theorem 1 reads off the halt.

4.2 Theorem 3: a monotone potential, no cycles

Theorem 3. $\Phi = a + b$ increases by at least 1 at every non-halting step. *Proof.* R4 (divide) sends $a + b = 3k + b$ to $k + (b + 2k + 1) = 3k + b + 1$, exactly +1. For the resets, b cancels: R5 gives $\Phi \rightarrow (4(a-1)/3 + b + 3) + 1$, an increment of $(a + 8)/3 > 0$; R6 gives $(a + 10)/3 > 0$; and the $a = 1$ exits R2/R3 give +3 and +2. The minimum, 1, is attained by divides. **QED** (verified on ~358,000 transitions across ten orders of magnitude, and 300,000 steps of the true orbit).

Corollary. The machine cannot cycle; from $A(1, 1)$ it halts or escapes to infinity. This is another affine potential in the collection — alongside machine 1's $\Phi = 2b + d$ — and it was found the same way: look for an affine combination in which the reset rules' fed-back variable cancels. The recipe now has a perfect record. As with the other two, this excludes the one scenario simulation can never rule out: an eventually-periodic orbit with an astronomically long transient (bbchallenge's Skelet #1).

5. Why it (almost certainly) never halts

Two independent rarities protect the machine. First, reaching $a = 1$ at all requires the reset stream to produce an **exact power of 3** — a measure-zero coincidence, and in the deep run the orbit never hit one (no $a = 1$ event in 3,000,000 composite steps). Second, even a power of 3 only halts if its exponent is divisible by 3, killing two thirds of the already-rare hits.

The branch statistic makes the rarity quantitative. The depth j of each divide chain is the 3-adic valuation of a reset value, and it follows the geometric law

$$P(j) = (2/3)(1/3)^{j-1} \quad (\text{measured: } 0.669, 0.219, 0.075, 0.025, \dots \text{ vs } 0.667, 0.222, 0.074, 0.025),$$

the base-3 analogue of the coin-flip behind the Collatz heuristic. The a -orbit grows geometrically (about 0.48 bits per reset, roughly $\times 4/3$ from the reset map net of the divide losses), so it passes the target scale 27^m after $O(m)$ resets. Modeling the reset values as arithmetically generic at their scale, the probability of landing exactly on 27^m decays geometrically in m , so the expected number of hits is a convergent sum — a **divergent-type cryptid** in the collection's dichotomy: geometric growth outrunning a geometrically sparse target, exactly like machine 1 and Space Needle. Non-halting is overwhelmingly likely; a proof is not available, for the usual single-orbit reason.

6. Status and verification

check	scope	result
implementation vs base rules	hand-derived prefix $A(1,1)..A(14,1)$; accelerated form matches base state-for-state on 177,717 composite steps of the real orbit	pass
Theorem 1 (halting criterion)	$a = 3^j$ halts iff $j \equiv 0 \pmod{3}$, for $j = 1..15$; halting set $\{27^m\}$	pass
Theorem 2 (divide-chain closed form)	$(N, b_0) \rightarrow (M, b_0 + (N-M) + j)$ on 30,000 constructed chains	pass
Theorem 3 (potential, no cycles)	$\Phi = a + b$ increment ≥ 1 on $\sim 358,000$ transitions (min 1) and 300,000 orbit steps	pass
branch statistic	divide depth = 3-adic valuation, $P(j) = (2/3)(1/3)^{j-1}$ to three decimals over 100,000 chains	pass
deep run	3,000,000 composite steps from $A(1,1)$, no halt; a reaches 708,337 bits; no power-of-3 ($a = 1$) event	pass

Artifacts: `m3_base.py` (the six rules), `m3_accel.py` (the composite step with batched divide-chains, self-verified against the base), `m3_theorems.py` (Theorems 1–3 and the measurements), `make_report3.py` (this report).

Status: open (divergent, multiplicative-coincidence cryptid); halts or escapes — periodicity excluded by proof.